RITHIK R NAMBIAR

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SUMMARY

MS Aerospace Engineering candidate at Iowa State University specializing in computational analysis and sustainable system design. Research portfolio spans offshore wind system controls, ultrasonic deterrent technologies, aerostat optimization, and computational aeroacoustics through collaborations with leading US and Indian institutions. Published researcher with proven track record of delivering practical engineering solutions. Passionate about translating complex engineering challenges into sustainable solutions that advance both technology and environmental stewardship.

EDUCATION

IOWA STATE UNIVERSITY **Aug 2024 - Aug 2026**

MS in Aerospace Engineering (with thesis)

MANIPAL INSTITUTE OF TECHNOLOGY **Jul 2018 - Jul 2022**

BTech in Aeronautical Engineering | Minor Specialization in Fundamentals of Computing

- Degree focused on aerodynamics, aircraft structures, flight mechanics, propulsion, and avionics.
- Recognized for the outstanding thesis work completed during the final semester in my batch.

EXPERIENCE

Department of Aerospace Engineering, Iowa State University

Graduate Research & Teaching Assistant **May 2025 – Present**

- Applying dynamic modeling and time-domain simulations for a novel Wave-Augmented Floating Offshore Wind Turbine (WAFOWT) system using MATLAB/Simulink and WEC-Sim to evaluate wave-body interactions and enhance platform stability.
- Serving as Teaching Assistant for AERE 3220: Aerospace Structures Laboratory (Fall 2026), guiding 30+ students through hands-on experiments to enhance understanding of aerospace structural principles.

Research Assistant, ISU Computational Aerodynamics and Aeroacoustics Laboratory **Sept 2024-Jan 2025**

- Designed a novel ultrasonic bat deterrent device that leverages aerodynamic whistles to mitigate wind turbine mortality risks at wind farms.
- Enhanced understanding of aerodynamic whistle performance by conducting detailed CFD and CAA simulations, identifying key operational principles and specific areas for design improvement.
- Applied Multidisciplinary Design Optimization (MDO) methods using Discrete Adjoint approach with OpenFOAM (DAFoam) to enhance device performance.

Indian Institute of Technology, Bombay, India

Project Research Assistant, LTA Systems Laboratory **Sept 2022-Jun 2024**

- Led feasibility study on solar/wind aerostats for transmission towers, reducing potential costs by 20%.
- Optimized power generation efficiency by 15% through CFD analysis of horizontal and vertical axis turbines.
- Developed dashboard to streamline aerostat design process, reducing cycle time by 60%.
- Guided students as a Teaching Assistant for AE664: LTA Systems, earning positive feedback for exceptional teaching support.
- Demonstrated leadership by organizing impactful events, outreach programs, guiding interns, and strengthening project visibility.

Indian Institute of Space Science and Technology, India

Research Intern – bachelor's thesis **Jan 2022-Jul 2022**

- Conducted collaborative research in my bachelor's thesis, employing ANSYS Fluent to dissect screech tones in supersonic jets.
- Evaluated sound pressure levels and acoustic behaviors across diverse microphone locations, providing valuable insights into aircraft performance.
- Explored the impact of Mach numbers on noise generation in supersonic jets, specifically focusing on predicting screech tones.

Hindustan Aeronautics Limited, Bangalore, India

Design Intern **Aug 2021-Sept 2021**

- Performed the design and analysis of critical aircraft components in the Wing and Empennage division, leading to heightened structural integrity and performance.
- Executed a focused study on gurney flaps, optimizing their application for a remarkable 15% improvement in lift coefficient, enhancing overall aerodynamics and maneuverability.

PUBLICATIONS

C-Conference proceeding | J – Journal publication | *-Extended abstract

2025

[J2] Nambiar, R. R, Karthik S. K, Manikandan, M., Singh, B., and Nair, M.T (2025) "Numerical Prediction of Axisymmetric Jet Screech Tones in Supersonic Flows."*CFD Letters*, 18(4), 34-53. https://doi.org/10.37934/cfdl.18.4.3453

2024

[J1] Nambiar, R. R, Dixit, M., & Pant, R. S. (2024). "Conceptual Sizing Approach for Electric Power Generation Turbine Aerostat." *Journal of Aerospace Sciences and Technologies*, *76*(2), 67–80. https://doi.org/10.61653/joast.v76i2.2024.962

2023

[C4] Nambiar, R. R, Tripathi, M., and Pant, R.S. "Effect of Horizontal and Vertical Overlap on the performance of two-bladed Vertical Axis Wind turbine." *AIAA AVIATION 2023 Forum*, San Diego, CA. June 2023. https://doi.org/10.2514/6.2023-4118

[C3] Nambiar, R. R, Dixit, M., and Pant, R. S. "Methodology for Conceptual sizing of a Turbine Aerostat for Electrical power generation." *AIAA AVIATION 2023 Forum*. San Diego, CA, June 2023. https://doi.org/10.2514/6.2023-3792

[C2] Nambiar, R. R, Karthik, S. K, Manikandan, M., and Nair, M.T. "Numerical Prediction of Generation of Axisymmetric Mode Jet Screech Tones at Different Mach numbers." *AIAA AVIATION 2023 Forum*, p. 4389. June 2023. https://doi.org/10.2514/6.2023-4389

[C1] Nambiar, R. R, and Gupta, P. "Conceptual Design of a Solar Powered Winged Hybrid Airship for Agricultural purposes." *AIAA AVIATION 2023 Forum*. San Diego, CA, June 2023. https://doi.org/10.2514/6.2023-3791

TECHNICAL SKILLS

- **Engineering Software**: Autodesk Fusion 360, CATIA v5, ANSYS, AutoCAD
- **Numerical Analysis Tools**: CFD++, MIME, ANSYS (Mechanical APDL, Fluent), OpenFOAM
- **Programming**: Python (Space Applications, Data Analysis), MATLAB
- **Documentation Tools**: Microsoft Office Suite, Google Workspace, LaTeX
- **Project Management:** Team Leadership, Cross-functional Collaboration, Research Coordination

TRAINING WORKSHOPS & SEMINARS

- Organized and successfully conducted a National Seminar on Emerging Technologies in Aerospace Engineering (eTAE – 2023) attended by dignitaries from the private and government agencies at IIT-Bombay on May'23
- Engaged a Program on Python fundamentals, Data Structures, Program Interfaces, Databases provided by University of Michigan through Coursera (online) in Sept'20
- Attended Coursework including 'hands-on projects that practiced skills in CAD, CAE, and Drone Design' provided by Coursera (online) in Oct'20

CERTIFICATIONS

- Simulation using ANSYS Fluent by Udemy (**Jun 2022)**
- Aircraft Design using CATIA V5 by Upskillz (**May 2022)**
- Python with Space Applications by Spaceonova (**Mar 2021)**
- Complete CATIA V5 Express Training by Udemy (**Jan 2021)**
- Python for Everybody (Specialization) by Coursera (**Oct 2020)**
- Autodesk Certified Credential in CAD and Digital Manufacturing (Specialization) by Coursera (**Sep 2020)**

OTHER

- **Languages**: English, Hindi, Malayalam (Fluent)